Reg. No. | | | | | | | | | | | [17AE44]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE–2019

AERODYNAMICS-I

Time:3 Hours **Maximum Marks:70**

Instructions: 1. Answer any **FIVE** full questions choosing at least **TWO** questions from each part.
2. Any missing data may be suitably assumed.

PART - A

1.	**a.**	Write down the expressions for the Navier stroke Equation and state the application of Navier stroke Equation.	**05**
	b.	Derive the Bernoulli's equation and state the assumptions made in derivation.	**09**
2.	**a.**	Explain all the Elementary flows with neat sketches.	**04**
	b.	Derive the radial and tangential velocities for the combination of uniform flow and source flow in terms of Velocity potential and Stream function.	**10**
3.	**a.**	Explain how Kelvin's circulation theorem helps in the generation of lift over an airfoil.	**05**
	b.	Derive the expression for complex potential function.	**09**
4.	**a.**	State the principle of Kutta condition with the help of a neat diagram.	**04**
	b.	Illustrate how to obtain flow past an airfoil from flow past a circular cylinder where radius is 1 using Joukowski transformation.	**10**

PART - B

5.	**a.**	**i.** Explain wing tip vortex with a neat sketch. **ii.** What is meant by Washin and Washout for wing?	**05**
	b.	Derive the fundamental equation for thin airfoil theory.	**09**
6.	**a.**	Write a short note on effect of geometrical parameters on lift and induced drag.	**06**
	b.	Define drag and with a neat sketch explain different types of drag.	**08**
7.	**a.**	Write a short note on boundary layer separation and airfoil stall.	**05**
	b.	For a finite wing derive the fundamental equation of Prandtl's lifting line theory.	**09**
8.	**a.**	With a neat sketch explain wing planform geometry.	**05**
	b.	Define shape factor. Explain and Write down its approximate values for laminar and turbulent boundary layers.	**09**

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